

Similarity

What is similarity?





AUSTRALIA

LEGEND

- State Boundary
- Major Road
- Rail
- River
- Major City
- Sea Port
- Country Capital
- Airport
- Point of Interest
- Salt Lake
- Mt. Peak

Australian Overseas Territories

- Norfolk Island
- Christmas Islands
- Heard and McDonald Islands
- Ashmore and Cartier Islands
- Cocos (Keeling) Islands
- Coral Sea Islands Territory
- Australian Antarctic Territory

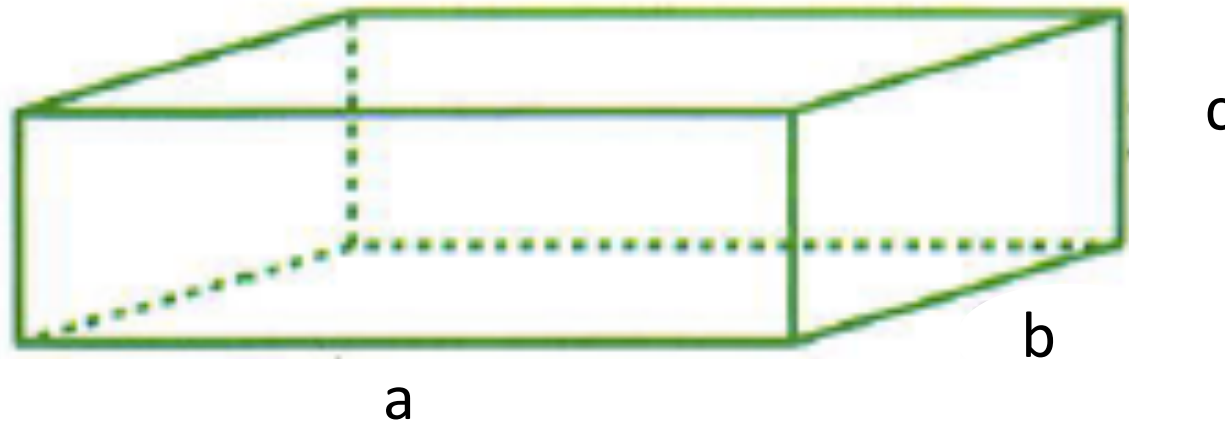
Map not to Scale

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(Updated on 17th May, 2013)

TASMANIA

(Updated on 17th May, 2013)

Calculate the surface area and volume for the rectangular prism (cuboid) below for the various dimensions listed.



Surface area = sum of the area of the flat surfaces
$$= 2 \times (a \times b + b \times c + a \times c)$$

Volume = $a \times b \times c$

Prism	Prism 1			Prism 2			Prism 3		
Dimension of the PRISM (m)	8	3	2	16	6	4	24	9	6
Surface area (m^2)	92			368			828		
Volume (m^3)	48			384			1296		

	Prisms 1 and 2	Prisms 1 and 3
Ratio of dimensions	1 : 2	1 : 3
Ratio of surface area	1 : 4	1 : 9
Ratio of volume	1 : 8	1 : 27

	Prisms 1 and 2	Prisms 1 and 3
Ratio of dimensions	1 : 2	1 : 3
Ratio of surface area	1 : 4 (1 : 2²)	1 : 9 (1 : 3²)
Ratio of volume	1 : 8 (1 : 2³)	1 : 27 (1 : 3³)

Prisms 2 and 3 is the enlargement of Prism 1. It can be summarised that

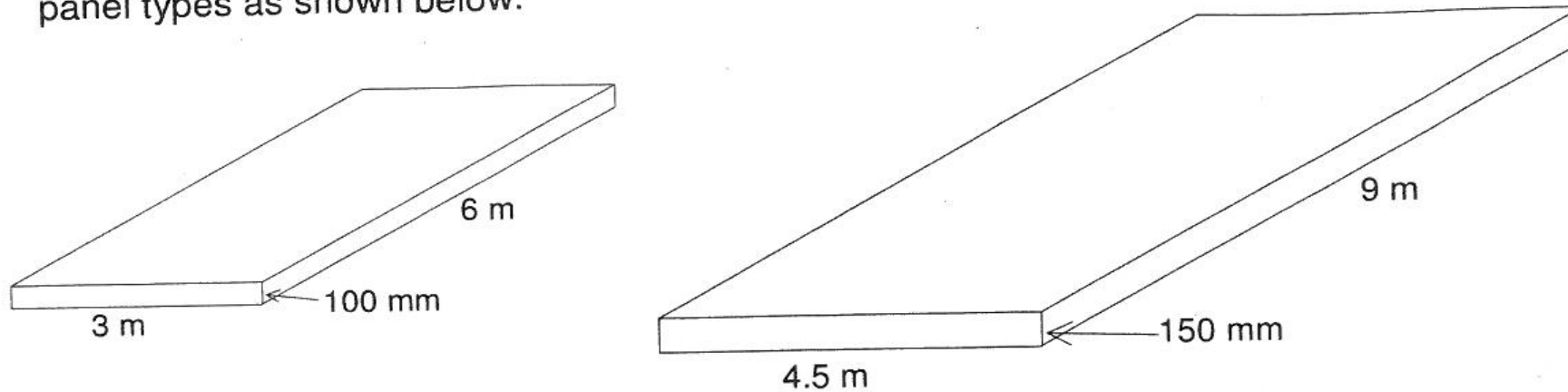
Lengths original : length in enlargement = 1 : k

Surface area (original) : surface area (enlargement) = 1 : k²

Volume (original) : volume (enlargement) = 1 : k³

k = the scale factor of enlargement

A manufacturer of precast concrete panels used in the building industry manufactures two similar panel types as shown below.



- (a) The cost of concrete to make the smaller panel is \$280. Calculate the cost of concrete to make the larger panel

Ratio of length smaller panel : larger panel = **3 : 4.5 = 1 : 1.5**

Cost of making smaller panel = \$280 (given)

To make the larger panel, the volume of concrete required is **1.5³** of the smaller panel.

Therefore the cost of making larger panel = **\$280 × 1.5³ = \$945.00**

Before being used the panels require one coat of a special sealant which costs \$35/L. One litre of this sealant covers an area of 3.5 square metres.

- (b) Calculate the cost of sealing the smaller panel.

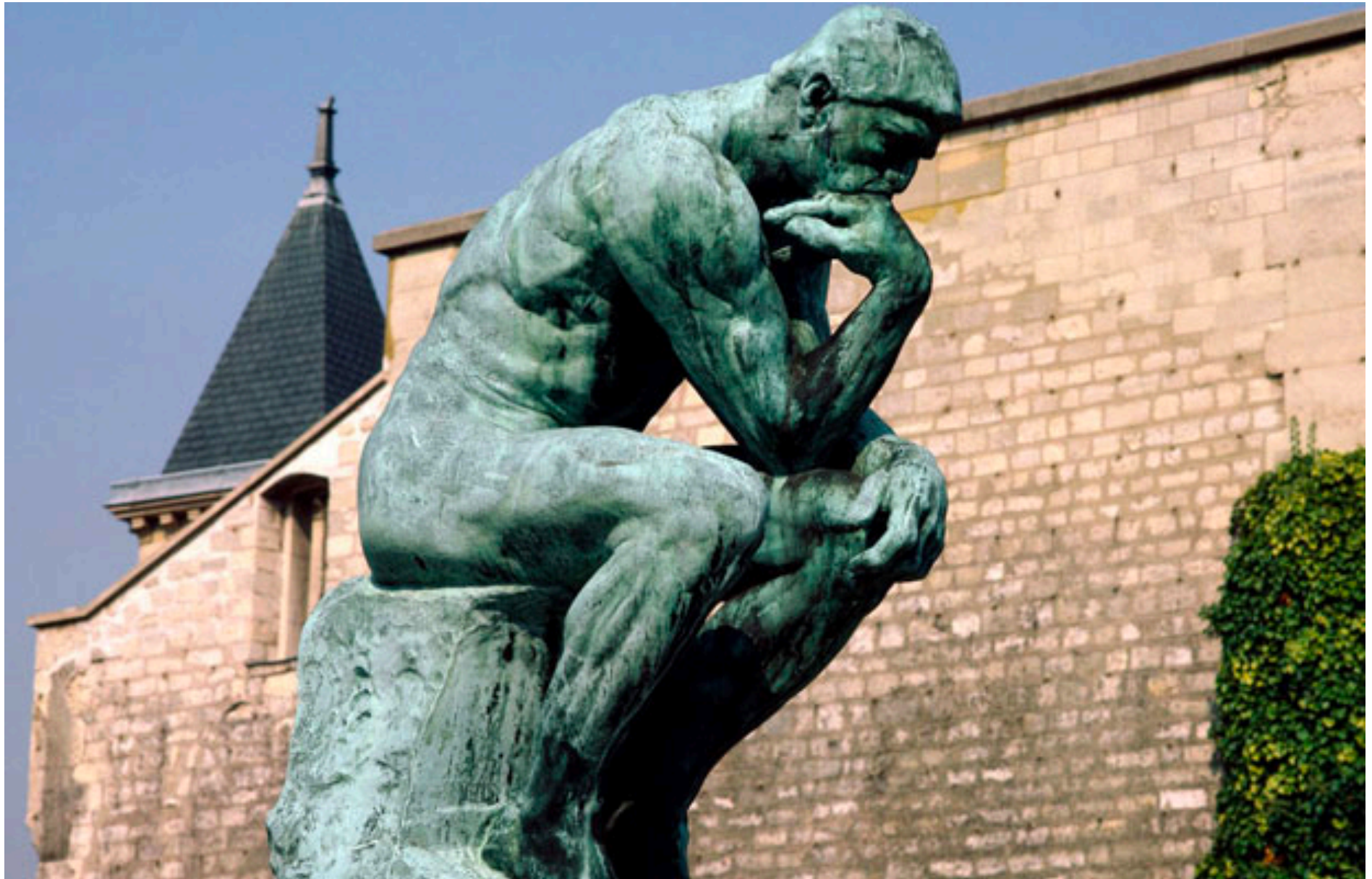
$$\text{Total surface area of the smaller panel} = 2 \times (3 \times 6 + 3 \times 0.1 + 6 \times 0.1) = \mathbf{37.8 \text{ m}^2}$$

$$\text{The cost of sealing the smaller panel} = \frac{37.8}{3.5} \times \$35 = \mathbf{\$378.00}$$

- (c) How much will it cost to seal the larger panel?

The surface area of the larger panel is **1.5²** of the smaller panel.

Therefore the cost to seal the larger panel = **\$378 × 1.5² = \$850.50**



When Emma was on holiday in Paris she purchase a replica of The Thinker, after visiting the Rodin Museum in Paris.

Her bronze replica was cast in the ratio 1:6. Emma's replica was 31 cm high and weighed 5 kg.

(a) Calculate the height of the sculpture.

(b) Estimate the weight of the sculpture.

(c) What is the ratio of the surface area of her replica to that of the sculpture?



a. Ratio of height – model : actual = 1 : 6 (given)

Actual height of The Thinker = $31 \times 6 = 186 \text{ cm}$

b. Weight is proportional to volume.

The ratio of volume – model : actual = $1 : 6^3$

The weight of The Thinker = $5 \times 6^3 = 1\,080 \text{ kg}$

c. Ratio of height – model : actual = 1 : 6 (given)

Therefore the ratio of surface area – model : actual = $1 : 6^2 = 1 : 36$

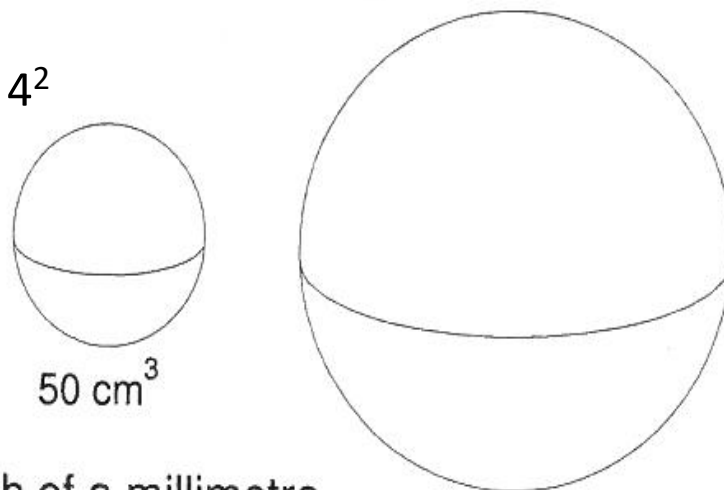
The ratio of surface areas of two spheres is 1 : 16, and the volume of the smaller sphere is 50cm^3 .

(a) What is the ratio of the diameters of these spheres?

Ratio of surface area – small : large = 1 : 16 (given) = $1 : 4^2$

Therefore the ratio of radius – small : large = **1 : 4**

(b) Calculate the volume of the larger sphere.



The volume of the larger sphere = $50 \times 4^3 = 3\,200\text{ cm}^3$

(c) Calculate the radius of the smaller sphere to the nearest tenth of a millimetre.

$$\text{Volume of the sphere} = 50\text{ cm}^3 = \frac{4}{3}\pi r^3$$

Use SOLVE function (classpad) to solve for r

$$r = 2.285\text{ cm} = 22.85\text{ mm} = \mathbf{22.9\text{ mm}}$$
 (to the nearest tenth of a mm)

(d) Calculate the surface area of the larger sphere to the nearest square centimetre.

$$\text{Surface area of the smaller sphere} = 4\pi r^2 = 4\pi (2.285)^2$$

$$\text{SA of the larger sphere} = 4\pi (2.285)^2 \times 16 = 1\,050\text{ cm}^2 \text{ (nearest square cm)}$$